

Purely Analytic Closure of the Final Spherical-Shell Residual

Abstract

This note assembles the all-frequency retained-remainder estimate after the exact owner subtraction. It proves that the final spherical-shell residual is empty without invoking an interval certificate, an executable truth table, an aggregate tail argument, or a frequency split. The structural, angular-block, and scalar-positivity modules cover one closed superset of the entire residual.

1 Notation and analytic inputs

Let

$$A_{\rho,1} := \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \quad W(\rho, K) := \frac{2(1-\rho^3)K^3}{9\pi},$$

and let $N_D(A_{\rho,1}, K^2)$ count Dirichlet eigenvalues strictly below K^2 . Put

$$\rho_c := \frac{1}{1+2\pi}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}.$$

The preceding analytic owner lemmas and the exact set subtraction give

$$\mathcal{D}_{20} = \left\{ (\rho, K) : \rho_c \leq \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho) \right\}. \quad (1)$$

On this interval $U(\rho) = K_0(\rho)$, although the formula for the upper wall will not be needed below.

We use the shell action

$$G_a(z) := \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right) \quad (0 \leq z < a), \quad G_a(z) := 0 \quad (z \geq a),$$

$$A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z),$$

and the strict lattice proxy

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu \# \left\{ n \in \mathbb{N} : n < A_{\rho,K}(\nu) + \frac{1}{4} \right\}. \quad (2)$$

The analytic Bessel-phase reduction proved earlier in the main argument is

$$N_D(A_{\rho,1}, K^2) \leq P(\rho, K). \quad (3)$$

Put

$$M_c(\rho, K) := \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}.$$

Parts VII–IX of the detailed analytic supplement establish, respectively, the following three inputs. The retained-remainder module gives the structural implication

$$\mathcal{E}_{\text{ang}}(\rho, K) < \mathcal{H}(\rho, K) \implies P(\rho, K) < W(\rho, K). \quad (4)$$

The paired angular-block module gives, on $\rho_c \leq \rho \leq 39/50$ and $K \geq k_{11}(\rho)$,

$$\mathcal{E}_{\text{ang}}(\rho, K) < M_c(\rho, K). \quad (5)$$

The scalar-positivity module gives on the same unbounded domain

$$\mathcal{H}(\rho, K) - M_c(\rho, K) > 0. \quad (6)$$

The all-frequency scope in (6) is part of that module: its scalar gap has positive K -derivative whenever $K > 12$, one has $k_{11}(\rho) > 241/20 > 12$, and the gap is strictly positive on the base curve $K = k_{11}(\rho)$. Thus the proof integrates upward from the base curve to every finite $K \geq k_{11}(\rho)$; it uses no upper-frequency bound.

2 All-frequency retained-remainder closure

Theorem 1 (All-frequency middle-ratio theorem). *For*

$$\rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho),$$

one has

$$N_D(A_{\rho,1}, K^2) < W(\rho, K).$$

Proof. Equations (5) and (6) give the strict chain

$$\mathcal{E}_{\text{ang}}(\rho, K) < M_c(\rho, K) < \mathcal{H}(\rho, K).$$

The structural implication (4) therefore gives $P(\rho, K) < W(\rho, K)$, and the phase reduction (3) completes the proof. Strictness includes both closed ratio faces, the base face $K = k_{11}(\rho)$, and every common radial–angular proxy wall. $\square$

3 Direct ownership of the exact residual

Proposition 2 (Analytic closure of $\mathcal{D}_{20}$). *Define the closed all-frequency theorem domain*

$$\mathcal{R}_{\text{tr}} := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho) \right\}.$$

Then $\mathcal{D}_{20} \subset \mathcal{R}_{\text{tr}}$, the strict Pólya inequality holds throughout $\mathcal{D}_{20}$, and the successor residual is empty.

Proof. If $(\rho, K) \in \mathcal{D}_{20}$, then (1) gives

$$\rho_c \leq \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho).$$

After weakening the strict upper ratio and lower frequency inequalities, these conditions place (ρ, K) in $\mathcal{R}_{\text{tr}}$. Therefore Theorem 1 proves every point of $\mathcal{D}_{20}$, and

$$\mathcal{D}_{20} \setminus \mathcal{R}_{\text{tr}} = \emptyset.$$

The already owned faces $\rho = 39/50$, $K = k_{11}(\rho)$, and $K = U(\rho)$ remain outside $\mathcal{D}_{20}$, even though the stronger theorem also covers the first two. The face $\rho = \rho_c$ remains in $\mathcal{D}_{20}$ whenever both strict frequency inequalities hold. No artificial frequency face is present. $\square$

Logical dependency. After Parts I–VI identify $\mathcal{D}_{20}$, the closure uses only the structural, angular-block, and scalar-positivity modules in Parts VII–IX. It does not use an aggregate-tail module, a frequency split, the former 10,580-box compact certificate, the 1,286-box aggregate certificate, or the 63-state executable subtraction table.